

Gabriel Mukobi

Web: gabrielmukobi.com | Email: gmukobi@cs.stanford.edu | Mobile: [360.525.7299](tel:360.525.7299) | GitHub: [mukobi](https://github.com/mukobi) | LinkedIn: [gabrielmukobi](https://www.linkedin.com/in/gabrielmukobi)

Summary:

Researcher, public servant, and leader passionate about research, policy, and strategy to secure a flourishing future with advanced AI systems. Experienced in AI research, software engineering, and leadership in both small-team and large-company environments.

Experience:

Agent Security Researcher, U.S. Center for AI Standards and Innovation - July 2024-Present - 40 h/wk - Washington, DC - nist.gov/caisi

Technical research to protect U.S. national security, public safety, and economic competitiveness from AI-enabled threats.

Research Fellow, RAND - Apr 2024-July 2024 - 40 h/wk - Washington, DC - rand.org

Informed and improved AI policy decision-making through technical and policy research and analysis.

Technical AI Research Fellow, ERA Fellowship - Jul 2023-Sep 2023 - 40 h/wk - Cambridge, UK - erafellowship.org

Built an evaluation of]cooperative AI capabilities with language models in multi-agent environments; also a Krueger Lab Intern.

Gameplay Engineering Intern, Respawn Entertainment - Jun 2022-Sep 2022 - 40 h/wk - Remote - ea.com

Engineered core gameplay and AI features on Respawn's unreleased Star Wars bounty hunter first-person shooter.

Gameplay Engineering Intern, Riot Games - Jun 2021-Sep 2021 - 40 h/wk - Remote - riotgames.com

Designed and implemented core features as a Software Engineering Intern on the gameplay team of [2XK0](https://2xk0.com) (prev. Project L).

Research Programmer Intern and Tools Programmer Intern, Epic Games - Jun 2020-Jan 2021 - 40 h/wk - Remote - unrealengine.com

Created deep reinforcement learning samples, a machine learning plugin, and virtual production tools in Unreal Engine.

Google Engineering Practicum Intern, Google Cloud Platform - Jun 2019-Sep 2019 - 40 h/wk - Seattle, WA - github.com/knative-portability

Developed full-stack [open-source applications](https://open-source-applications.com) as proof of portability for [Knative](https://knative.dev), a platform for serverless containerized workloads.

Selected Projects:

Escalation Risks from Language Models in Military and Diplomatic Decision-Making - Oct 2023-Jan 2024 - [Paper](#), [GitHub](#)

Co-first author. Evaluating the risks from autonomous language model decision-makers in escalating international conflicts. Accepted to ACM FAccT 2024, MASEC NeurIPS 2023 workshop (spotlight).

Welfare Diplomacy: Benchmarking Language Model Cooperation - Jun 2023-Sep 2023 - [Paper](#), [GitHub](#)

First author. Multi-agent LLM evaluations in a novel general-sum variant of Diplomacy that better incentivizes and measures cooperation. Accepted to the SoLaR NeurIPS 2023 workshop, in review at ICML 2024.

SuperHF: Supervised Iterative Learning from Human Feedback - Jan 2023-Sep 2023 - [Paper](#), [GitHub](#)

First author. Alternative to RLHF using supervised learning instead of RL. Accepted to the SoLaR NeurIPS 2023 workshop.

Societal Adaptation to Advanced AI - Jan 2024-May 2024 - [Paper](#), [Thread](#)

Co-first author. How society might adapt to possible future risks from advanced AI systems.

Skills:

Artificial Intelligence - software.gabrielmukobi.com/ai

AI security, NLP, AI governance, evaluations, ML, DL, foundation models, research mentorship. Languages: Python.

Software Engineering - software.gabrielmukobi.com

Product management, documentation, testing, bug reporting, code review, CS, VCS, [GitHub](#), [GitLab](#). Languages: Python, C++, C#.

Web Development - software.gabrielmukobi.com/web

Full-stack, web design, cloud computing, databases, Docker containerization. Languages: JavaScript, Node.js, Python, HTML.

Game Development - software.gabrielmukobi.com/games

Unreal Engine, Unity, gameplay programming, tools, virtual reality, 3D modeling, computer graphics. Languages: C++, C#, Python.

Education:

University of California, Berkeley - Ph.D. Computer Science - Aug 2024-Leave

Ph.D. student on leave researching machine learning security. Advised by [Jacob Steinhardt](#) and [Dawn Song](#).

Stanford University - M.S. Computer Science - Sep 2023-Mar 2024, B.S. Computer Science - Sep 2018-Dec 2023 - Cum GPA: 4.01

Coursework and extracurriculars in AI/ML, Graphics, Computer Systems, Algorithms, and Theory.